Sameer Baskota

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Lalitpur, Nepal

RESEARCH INTERESTS

Unmanned Aerial Vehicles (UAVs), agricultural robotics, autonomous flight systems, and AI-driven aerial imaging. Interested in UAV design, system integration, flight operations, and the application of computer vision and machine learning to precision agriculture, crop monitoring, and real-world agricultural challenges.

EDUCATION

IOE, Pulchowk Campus 2022 – 2026
Bachelor in Aerospace Engineering - Location: Lalitpur, Nepal

EXPERIENCE

Research Assistant - Geovation Nepal Jul 2026 – Present
- Conduct field operations for drone-based precision seeding in agricultural fields.
- Investigate factors affecting seeding efficiency, including payload capacity, flight parameters, and seed distribution mechanisms
- Study the challenges and opportunities of adopting agricultural drones in Nepal.
- Investigate ways to improve the performance, reliability, and efficiency of agricultural drones for use in Nepal.

Internship - AMO (Yeti Airlines) Nov 2025 – Jan 2026
- Supported aircraft maintenance operations through coordination with MCC and AMO teams
- Assisted in maintenance documentation, tooling, and spare-parts inventory management
- Participated in Quality Assurance and production-planning related audit activities

Flying Club, Pulchowk Campus Dec 2024 – Jan 2025
- Participated in UAV design, fabrication, and flight testing
- Gained hands-on experience with RC aircraft assembly, drone systems, and flight operations
- Worked with a multidisciplinary team on aerodynamic design and flight control systems

PROJECTS

UAV-Based Crop Health Monitoring Using RGB Based Imaging Aug 2025 – Mar 2026
Tools: Python, OpenCV, DJI Phantom 4, QGIS, DroneDeploy
- Developed a UAV-based system for maize crop health monitoring
- Generated orthomosaic maps from RGB aerial imagery
- Extracted vegetation indices (ExG, VARI, and NGRDI) for analysis
- Built a Random Forest model for nutrient-deficiency classification

Research Paper Ranking And Recommendation System (SIC Project) May 2025
Tools: Python, SciBERT, SBERT, OpenCV
- Developed an AI-based research paper ranking and recommendation system
- Integrated OCR and computer vision for figure and table extraction
- Built a web interface for interactive literature analysis
- Applied NLP models (SciBERT and SBERT) for paper relevance evaluation

Adaptive Drone Swarm Behavior Simulation Jan 2025
Tools: Python, NumPy, Matplotlib, Multi-Agent Simulation
- Developed a simulation framework for adaptive drone swarm behavior
- Implemented coordinated movement and formation control among multiple drones
- Designed obstacle-avoidance and dynamic path-planning algorithms
- Evaluated swarm performance under varying environmental conditions

Aerombulance – Ambulance Aircraft for Medical Operations July 2025
Tools: CATIA, XFLR5, Plane Maker, X-Plane
- Designed a STOL (Short Takeoff And Landing) aircraft for remote and short-field operations
- Developed a modular cabin for patient and neonatal transport
- Defined mission profiles and system requirements for medical operations

ADDITIONAL PROJECTS

Autonomous UAV Navigation Using GPS and Pixhawk

- Assembled a quadcopter using a Pixhawk flight controller
- Integrated GPS and configured autonomous waypoint missions
- Calibrated onboard sensors and flight-control parameters
- Performed test flights and basic flight performance evaluation

UAV Mapping and Orthomosaic Generation

- Captured aerial imagery for UAV-based mapping applications
- Generated orthomosaic maps from drone image datasets
- Evaluated image overlap and survey coverage for mapping accuracy

Real Time Face Detection on Tello Drone

Tools: Python, OpenCV, DJI Tello SDK

- Developed a real-time face detection system using OpenCV and Python
- Integrated the system with the DJI Tello video stream
- Tested live face detection during drone flights

RC Fixed Wing Design and Flight Testing

- Designed and fabricated an RC fixed-wing aircraft for flight testing
- Integrated propulsion, control surfaces, and radio communication systems
- Carried out test flights to check flight performance
- Observed stability and control behavior during flight tests

Drone Payload Dropping Mechanism

- Designed a lightweight payload release mechanism for a quadcopter
- Integrated a servo-based actuation system for payload deployment
- Tested the mechanism for controlled release during flight

SKILLS

Programming: Python, MATLAB, C, Arduino

Machine Learning & AI: OpenCV, Scikit-learn, Random Forest, Computer Vision, Image Segmentation, Feature Extraction, NLP (SciBERT, SBERT)

UAV & Aerospace: Pixhawk Flight Controller, Autonomous Flight Setup, UAV Mapping, RC Plane Design

Design & Mapping Software: CATIA, QGIS, DroneDeploy

Research Skills: Data Collection, Data Analysis, Literature Review, Technical Report Writing (LaTeX)

TRAININGS & CERTIFICATIONS

- Samsung Innovation Campus (SIC) Python Training
- CAN-SAT Training, Orion Space
- Work in Machine Learning for Agriculture Applications

EXTRA CURRICULAR

- Event Director, The AeroHour Program, Pulchowk Campus
- Participant, Robo Soccer Competition, Mechtrix
- Program Coordinator, Boat Racing Competition, Pulchowk Campus
- First Position, STEAM Exhibition, Attariya , Kailali

REFERENCES

Asst. Prof. Kamal Darlami
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